SABER: Support-Conditioned Adaptive Bellman Error Reweighting for Sparse HER Actor-Critic

Anonymous submission

Abstract

Hindsight Experience Replay (HER) converts failed experience into goal-conditioned training data through achieved-goal relabeling. Most HER extensions alter goal selection, replay selection, or target construction. We instead ask whether per-sample squared Bellman residuals should remain equally weighted once relabeling and replay sampling are fixed. We introduce SABER (Support-Conditioned Adaptive Bellman Error Reweighting), a critic-only objective. SABER upweights a sample's squared Bellman residual when a sufficiently populated local replay set from the same discretized state–goal–horizon context contains an action in the current-policy action bin. It normalizes the weights to unit minibatch mean and uses the current-policy action only for the support query while retaining Bellman regression at the replay action. HER relabeling, uniform replay sampling, the actor objective, Bellman targets, network architectures, and the update-to-data (UTD) ratio remain unchanged. Under matched UTD-1 training on FetchPickAndPlace, SABER raises mean normalized success AUC from 0.539 to 0.609 with DDPG, from 0.372 to 0.442 with TD3, and from 0.471 to 0.583 with SAC. Paired-bootstrap 95% confidence intervals for the AUC differences exclude zero for TD3 and SAC but include zero for DDPG. On DrawerOpen with DDPG, SABER raises mean AUC from 0.207 to 0.754 and mean final success from 0.390 to 1.000. Controlled comparisons, ablations, and diagnostics examine alternative allocation mechanisms and characterize SABER's support signal. These gains, achieved under the same one-update budget and an otherwise unchanged HER pipeline, establish post-HER critic-loss allocation as a distinct design axis capable of substantial performance improvements.

1 Introduction

A central challenge in sparse-reward goal-conditioned reinforcement learning is obtaining useful supervision when successful trajectories are rare. Hindsight Experience Replay (HER) addresses this challenge by relabeling failed experience with achieved goals (Andrychowicz et al. 2017). Once sampling and relabeling are fixed, however, standard HER actor–critics assign equal coefficients to all sampled Bellman residuals. This leaves a separate question: should critic supervision remain uniform when the critic regresses at replay actions but the actor queries the critic at current-policy actions?

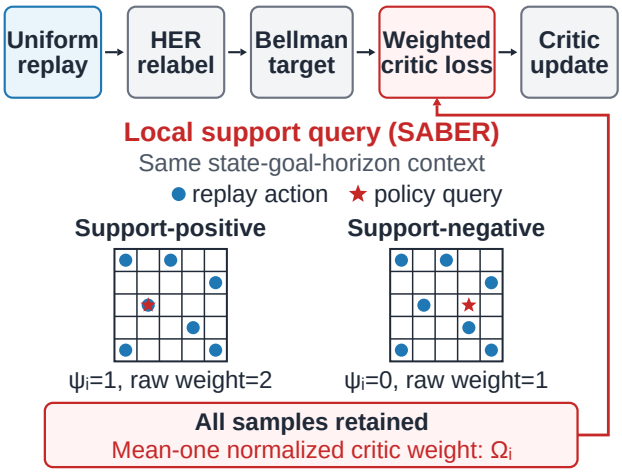


Figure 1: SABER reweights post-HER critic residuals using same-context replay support for the current-policy action; all samples are retained.

We introduce SABER (Support-Conditioned Adaptive Bellman Error Reweighting), a critic-only post-HER objective. SABER tests whether replay actions from the same discretized state–goal–horizon context occupy the current-policy action bin, upweights support-positive residuals, and normalizes weights to unit minibatch mean. The current-policy action is used only for this support query; regression remains at the replay action. Relabeling, replay sampling, rewards, Bellman targets, actor optimization, architectures, and the update budget remain unchanged.

Our contributions are: (i) identifying post-HER critic-loss allocation as distinct from goal selection, replay prioritization, and target construction; (ii) introducing a normalized local-support objective that preserves the underlying HER actor–critic pipeline; and (iii) evaluating its cross-backbone repeatability, query and count-matched controls, update-budget trade-offs, diagnostics, and task-dependent success and failure boundaries.

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

2 Related Work

Universal value function approximators formalize generalization across goals (Schaul et al. 2015), while recent theory characterizes sample-efficiency gains from goal relabeling (Zheng et al. 2024). Around HER, methods modify several stages. Hindsight Goal Generation (HGG) generates goals that balance short-term reachability with progress toward the desired goal (Ren et al. 2019). Curriculum-guided Hindsight Experience Replay (CHER) selects hindsight experiences through a curriculum based on goal proximity and pseudo-goal diversity (Fang et al. 2019). Energy-Based Prioritization (EBP) prioritizes episodes by object-trajectory energy (Zhao and Tresp 2018). Hindsight Goal Ranking (HGR) instead prioritizes both episodes and transition–future-goal pairs by temporal-difference (TD) error with two-stage importance correction (Luu and Yoo 2021). Multi-step Hindsight Experience Replay (MHER) uses multi-step relabeled returns (Yang et al. 2023), Aggressive Rewards to Counter Bias in Hindsight Experience Replay (ARCHER) rescales hindsight rewards (Lanka and Wu 2018), and Unbiased Sampling for Hindsight Experience Replay (USHER) corrects hindsight-induced value bias in stochastic environments (Schramm et al. 2023).

General replay methods redistribute experience independently of HER. Prioritized Experience Replay (PER) samples by TD error with importance correction (Schaul et al. 2016). Under suitable transformations, non-uniform sampling and a uniformly sampled loss can induce the same expected gradient (Fujimoto, Meger, and Precup 2020). Regret Minimization Experience Replay further derives priorities from Bellman error, on-policiness, and value accuracy (Liu et al. 2021). Attentive Experience Replay (AER) selects transitions according to state similarity with the agent's current state (Sun, Zhou, and Li 2020). Remember and Forget for Experience Replay (ReF-ER) suppresses far-policy samples while regularizing the actor toward replay behavior (Novati and Koumoutsakos 2019).

Action support is also central in offline RL: BCQ restricts policy actions toward the batch distribution, BEAR constrains backup actions through support matching, and CQL conservatively regularizes values on out-of-distribution actions (Fujimoto, Meger, and Precup 2019; Kumar et al. 2019, 2020). SABER instead trains online with HER, constrains neither the actor nor Bellman target, and uses local support only to allocate coefficients among replay-action residuals.

Closer critic-reweighting methods learn different signals. Experience Replay with Likelihood-free Importance Weights estimates a current-policy-to-replay density ratio (Sinha et al. 2022). Distribution Correction (DisCor) reweights samples according to estimated target accuracy (Kumar, Gupta, and Levine 2020). ReenGAGE adds gradient-based knowledge distillation to goal-conditioned critic learning (Levine and Feizi 2023). Across these lines, SABER differs in its implemented intervention and support signal. After HER relabeling and replay sampling are fixed, it preserves minibatch membership, rewards, targets, and the actor objective, and uses a non-learned, sample-aligned current-policy action-support predicate only to set critic-loss coefficients. It uses no density ratio, target-error model, or auxiliary target.

3 SABER

3.1 Problem Formulation and the Post-HER Critic-Loss Allocation Gap

We consider a finite-horizon goal-conditioned Markov decision process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{G}, P, r, \gamma, H)$. At time t, the environment is in state $s_t \in \mathcal{S}$, the agent receives observation $o_t = \omega(s_t)$, and the desired goal is $g \in \mathcal{G}$. The goal-conditioned input is $x_t = (o_t, g)$, and the policy and critic are denoted by $\pi_\theta(a \mid x)$ and $Q_{\phi_k}(x, a)$, respectively. Let $m : \mathcal{S} \to \mathcal{G}$ map a state to its achieved goal. The environment provides the sparse goal-conditioned reward

$$r(s_t, a_t, s_{t+1}, g) = \begin{cases} 0, & d(m(s_{t+1}), g) \le \epsilon_g, \\ -1, & \text{otherwise.} \end{cases} \tag{1}$$

When successful trajectories are rare, replay collected under the original desired goal contains little direct successful supervision.

Let an episode contain states $s_0, \ldots, s_T$ and transitions indexed by $t = 0, \ldots, T-1$. In our future-goal HER implementation (Andrychowicz et al. 2017), a transition selected for relabeling samples a later stored transition u and uses its next achieved goal:

$$\begin{aligned} u &\sim \mathrm{Unif}\{t+1, \ldots, T-1\}, \\ \tilde{g}_i &= m(s_{u+1}), \\ \tilde{r}_i &= r(s_t, a_t, s_{t+1}, \tilde{g}_i). \end{aligned} \tag{2}$$

When the eligible set is empty, or when a transition is not selected for relabeling, the original desired goal is retained. The resulting network inputs are $x_i = (o_i, \tilde{g}_i)$ and $x'_i = (o'_i, \tilde{g}_i)$. For backbone A, let y_i^{A} denote its standard Bellman target after relabeling and reward recomputation. For critic k, the replay-action Bellman residual is

$$\delta_{ik} = Q_{\phi_k}(x_i, a_i^r) - y_i^{\mathsf{A}}, \tag{3}$$

where a_i^r is the action stored in replay. Given a uniformly sampled and relabeled minibatch B, the standard HER critic objective is

$$\mathcal{L}_Q^{\mathrm{HER}} = \frac{1}{|B|} \sum_{i \in B} \sum_{k=1}^{n_Q} \delta_{ik}^2. \tag{4}$$

Thus, once replay sampling and relabeling are fixed, every sampled residual receives the same explicit coefficient.

HER determines which relabeled goals and rewards are used for learning, but it does not determine how critic fitting effort should be distributed among the residuals in the resulting minibatch. This gap may matter in off-policy actor–critic learning: the critic regresses Bellman targets at replay actions a_i^r, whereas the actor improves the policy by querying the critic at current-policy actions a_i^π. The amount of replay evidence near these policy queries can vary across observation-derived state–goal contexts. We therefore ask whether observable local replay support for a_i^π can be used to allocate critic-loss weight while holding the relabeled minibatch and Bellman targets fixed.

Because this question requires both replay evidence and a replay-based critic objective, SABER is designed for

replay-based off-policy actor–critic learning. We instantiate it with three representative continuous-control backbones: DDPG (Lillicrap et al. 2016), which uses a deterministic actor and one critic; TD3 (Fujimoto, van Hoof, and Meger 2018), which uses a deterministic actor, twin critics, target-policy smoothing, and delayed actor updates; and SAC, which uses a stochastic entropy-regularized actor and twin critics (Haarnoja et al. 2018). These algorithms provide representative deterministic and stochastic off-policy structures while retaining the replay-based critic update required by SABER. SABER leaves each backbone's Bellman target, bootstrap convention, and actor objective unchanged.

3.2 SABER Overview and Intervention Point

SABER is a per-sample critic-loss objective, not a network layer or replay sampler. As Fig. 1 shows, it acts after uniform replay, HER relabeling, reward recomputation, and target construction but before critic-loss reduction. For each relabeled sample, it queries whether the current-policy action bin is occupied by same-context replay actions and applies the resulting normalized weight to the Bellman residual at the original replay action. All sampled transitions, relabeled goals, rewards, and targets remain unchanged.

3.3 Current-Policy Local Support

For sample i, SABER constructs a task-specific discrete state–goal–horizon context from the agent-observed input, represented by the full support-context key

$$c_i = C(o_i, \tilde{g}_i, h_i), \tag{5}$$

where h_i is the remaining horizon. Here, "state" refers only to task-relevant state features derived from o_i, not to privileged simulator state s_i. In the manipulation domains evaluated in Sec. 4, the full key summarizes task progression, agent–object interaction, goal-relative geometry, gripper state, and remaining horizon. Concretely, FetchPickAndPlace uses manipulation phase, hand–object distance, object–goal distance, object height, gripper gap, and remaining horizon. The Drawer tasks use interaction phase, hand–handle distance, handle–goal distance, handle position along the opening axis, gripper state, and remaining horizon. These features are discretized using fixed, task-specific bins.

To bound lookup cost, SABER first uses a lower-dimensional *coarse retrieval key* $C^{\mathrm{ret}}(o, h)$, constructed from a subset of the full-key components, only to retrieve candidate replay slots. Let $\mathbf{J}_i^{\mathrm{ret}}$ denote the ordered list of currently valid, unique slots returned by the maintained index for the query's coarse retrieval key. Invalid entries and duplicate slot identifiers are removed in index order, retaining the first valid occurrence. SABER examines at most N_{scan} candidates:

$$S_i = \mathrm{Tail}_{N_{\mathrm{scan}}}\left(\mathbf{J}_i^{\mathrm{ret}}\right), \qquad |S_i| \le N_{\mathrm{scan}}. \tag{6}$$

Here the tail is taken in the index's deterministic maintained order and is not interpreted as temporal recency. The coarse retrieval key is an indexing device, not the support criterion. Candidates are filtered using the full support-context key:

$$I_i = \{j \in S_i : C(o_j, \tilde{g}_i, h_j) = C(o_i, \tilde{g}_i, h_i)\}. \tag{7}$$

This equality is exact component-wise matching: a candidate is retained only when every discretized component of its full key equals the corresponding query component. The relabeled query goal $\tilde{g}_i$ is used to compute both sides of this comparison, so a candidate is not matched using its originally stored desired goal.

Each action dimension is normalized by its environment bounds and divided into K bins. Let $b_K(a)$ denote the resulting multidimensional action bin and

$$\mathcal{U}_i = \left\{b_K(a_j^r) \mid j \in I_i\right\} \tag{8}$$

the set of replay-action bins occupied in the local context. With $n_i = |I_i|$, SABER defines the support indicator

$$\psi_i = \mathbf{1}\{n_i \ge N_{\mathrm{local}}\}\, \mathbf{1}\{b_K(a_i^\pi) \in \mathcal{U}_i\}. \tag{9}$$

The first condition requires a sufficiently populated local context; the second tests whether the current-policy action bin has been observed in that context. The main setting uses

$$K = 5, \qquad N_{\mathrm{scan}} = 512, \qquad N_{\mathrm{local}} = 4. \tag{10}$$

DDPG and TD3 use their deterministic actor action as a_i^π. For the support query, SAC uses the deterministic squashed mean of its actor while retaining stochastic actions in its standard actor and target objectives. The indicator uses neither Q-values nor TD errors. It is a binary occupancy test, and the current-policy action is not substituted for a_i^r in the Bellman residual.

3.4 Normalized Critic-Only Objective and Backbone Integration

SABER assigns the raw per-sample weight

$$w_i^{\mathrm{raw}} = 1 + \lambda_S \psi_i \tag{11}$$

and normalizes the weights by their minibatch mean:

$$\bar{w}_B = \frac{1}{|B|}\sum_{j\in B} w_j^{\mathrm{raw}}, \qquad \Omega_i = \frac{w_i^{\mathrm{raw}}}{\max(\epsilon_w, \bar{w}_B)}. \tag{12}$$

Here $\epsilon_w = 10^{-8}$ is a numerical safeguard; it is inactive in all evaluated variants because their raw weights are at least one. The main method uses $\lambda_S = 1$, so support-positive and support-negative samples receive raw weights two and one, respectively. If $p_B = |B|^{-1}\sum_{i\in B}\psi_i$ is the support-positive fraction, then

$$\Omega_i = \begin{cases} \dfrac{1+\lambda_S}{1+\lambda_S p_B}, & \psi_i = 1, \\ \dfrac{1}{1+\lambda_S p_B}, & \psi_i = 0. \end{cases} \tag{13}$$

This makes the allocation effect explicit: positive samples receive greater relative weight, negative samples remain in the update with positive weight, and the minibatch mean coefficient is one. If all samples share the same indicator value, normalization returns $\Omega_i = 1$ for every sample.

The SABER critic objective is

$$\mathcal{L}_Q^{\mathrm{SABER}} = \frac{1}{|B|}\sum_{i\in B}\Omega_i \sum_{k=1}^{n_Q}\delta_{ik}^2. \tag{14}$$

We use $n_Q = 1$ for DDPG and $n_Q = 2$ for TD3 and SAC; for twin-critic backbones, the same Ω_i is applied to both squared residuals. SABER therefore changes the relative coefficients of existing residual-gradient terms without adding samples, targets, or auxiliary losses. Mean-one normalization controls the average coefficient but does not assert that the numerical loss or gradient norm is unchanged, because weights may correlate with residual magnitude.

The main variant is critic-only:

$$\mathcal{L}_\pi^{\mathrm{SABER}} = \mathcal{L}_\pi^{\mathrm{A}}. \tag{15}$$

We use critic-only weighting to isolate post-HER Bellman-loss allocation; actor weighting is possible but would alter a second learning objective. DDPG, TD3, and SAC therefore retain their original actor-update rules, without implying that critic-only weighting is performance-optimal.

One SABER update therefore proceeds as follows: sample replay transitions uniformly; apply future-goal HER and recompute rewards; construct the backbone's unchanged Bellman targets; query current-policy local support; compute ψ_i, w_i^{raw}, and Ω_i; update the critic with $\mathcal{L}_Q^{\mathrm{SABER}}$; and perform the backbone's original actor and target-network updates. HER probability, replay sampling, network architecture, optimizer, exploration process, Bellman targets, and UTD remain unchanged.

Finally, support positivity is a binary occupancy proxy, not an estimate of action density, uncertainty, target accuracy, or target variance. SABER does not regress the critic at a_i^π, change exploration, or discover useful actions absent from replay.

4 Experiments

4.1 Common Experimental Protocol

Tasks, cohorts, and training protocol. We evaluate FetchPickAndPlace-v4 from the multi-goal robotics suite (Plappert et al. 2018) (FPP; horizon 50, 1M environment steps) with DDPG, TD3, and SAC, and the Meta-World (Yu et al. 2020) DrawerOpen and DrawerClose tasks (horizon 200, 2M steps) with DDPG and TD3. Primary comparisons use paired seeds 1–5, the future-goal relabeling convention defined in Section 3 with nominal relabeling probability $p_{\mathrm{HER}} = 0.8$, and UTD 1. Matched HER–SABER cohorts use identical replay and relabeling code, so their intended difference remains the per-sample critic-loss coefficients. We make no paired claims for SAC on the Drawer tasks because no complete matched HER–SABER cohort exists: the available DrawerOpen SAC+HER runs remain at zero success for all five seeds, and neither matched SABER runs nor DrawerClose SAC results were collected. Sparse rewards follow HER (Andrychowicz et al. 2017); backbone targets and policy updates follow DDPG (Lillicrap et al. 2016), TD3 (Fujimoto, van Hoof, and Meger 2018), and SAC (Haarnoja et al. 2018). Shared continuous-control settings are fixed within each matched task–backbone cohort without method-specific retuning; complete values, implementation details, and provenance are provided in the supplement.

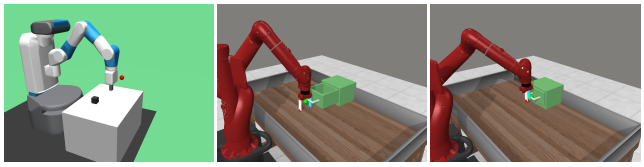


Figure 2: Illustrative simulator renderings of the experimental tasks. The panels are not evaluation rollouts.

Evaluation and statistics. Policies are evaluated deterministically over 20 episodes at each checkpoint. Paired analyses use only exact shared evaluation steps, without smoothing, interpolation, or filling. Normalized success AUC over the complete shared evaluation span is the primary metric. Final-quarter success AUC is a secondary late-training summary: it applies the same normalized trapezoidal calculation to the final 25% of the environment-step budget, namely 750k–1M on FPP and 1.5M–2M on the Drawer tasks. Final success is also secondary; Hit90 (the first checkpoint with success ≥ 0.9, with reached-seed coverage) is reported in the supplement. We report seed means and sample standard deviations; paired claims use within-seed differences and 10,000-resample percentile-bootstrap 95% confidence intervals. Absolute AUC values are not compared across tasks with different horizons or budgets.

4.2 Primary Comparisons

Comparison setup. HER addresses sparse supervision through goal relabeling, but learning effort can be reallocated at several stages of the resulting training pipeline. We compare methods that intervene at three operationally distinct points. HGR changes hindsight-experience selection by prioritizing episodes and transition–future-goal pairs according to TD error, with importance correction at both stages (Luu and Yoo 2021). PER changes replay selection by sampling replay transitions according to TD-error priorities and applying annealed importance correction (Schaul et al. 2016). SABER instead preserves both selections and changes only the critic-loss coefficients within the sampled and relabeled minibatch. These methods are therefore alternative interventions within the HER actor–critic pipeline rather than variants of a single mechanism.

Additional validated DDPG and SAC controls comparing support-based weighting with TD-error-based critic-loss weighting under uniform replay are reported in the supplementary material. AER (Sun, Zhou, and Li 2020), high-replay-ratio HER, and REDQ-HER variants based on REDQ (Chen et al. 2021; Hiraoka 2023) are also retained there as secondary scaling references because they additionally change HER usage, update budget, or architecture.

Table 1 summarizes final success, full-span success AUC, and final-quarter success AUC across all task–backbone cohorts. Hit90 and reached-seed coverage are reported in the supplement. Figure 3 complements these aggregate metrics by showing FPP learning dynamics across all three backbones.

Task	Backbone	Method	Final	AUC	FQ-AUC
FPP	DDPG	**SABER (Ours)**	$\mathbf{0.960 \pm .042}$	$\mathbf{0.609 \pm .038}$	$\mathbf{0.961 \pm .014}$
		HER	$0.940 \pm .042$	$0.539 \pm .125$	$0.910 \pm .101$
		HGR	$0.540 \pm .365$	$0.338 \pm .149$	$0.481 \pm .258$
		PER	$0.780 \pm .438$	$0.460 \pm .238$	$0.766 \pm .403$
	TD3	**SABER (Ours)**	$\mathbf{0.970 \pm .045}$	$\mathbf{0.442 \pm .035}$	$\mathbf{0.804 \pm .103}$
		HER	$0.840 \pm .188$	$0.372 \pm .032$	$0.607 \pm .076$
		HGR	$0.610 \pm .198$	$0.367 \pm .034$	$0.534 \pm .049$
		PER	$0.660 \pm .204$	$0.369 \pm .034$	$0.562 \pm .074$
	SAC	**SABER (Ours)**	$\mathbf{0.990 \pm .022}$	$0.583 \pm .030$	$\mathbf{0.986 \pm .011}$
		HER	$0.790 \pm .288$	$0.471 \pm .079$	$0.735 \pm .247$
		HGR	$0.760 \pm .251$	$0.484 \pm .071$	$0.664 \pm .217$
		PER	$\mathbf{0.990 \pm .022}$	$\mathbf{0.690 \pm .046}$	$\mathbf{0.986 \pm .011}$
DrawerOpen	DDPG	**SABER (Ours)**	$\mathbf{1.000 \pm .000}$	$0.754 \pm .044$	$\mathbf{0.987 \pm .018}$
		HER	$0.390 \pm .534$	$0.207 \pm .287$	$0.393 \pm .538$
		HGR	$0.900 \pm .224$	$0.688 \pm .142$	$0.982 \pm .027$
		PER	$0.990 \pm .022$	$\mathbf{0.825 \pm .044}$	$0.982 \pm .021$
	TD3	**SABER (Ours)**	$0.590 \pm .539$	$0.362 \pm .338$	$\mathbf{0.598 \pm .546}$
		HER	$0.190 \pm .425$	$0.127 \pm .283$	$0.200 \pm .447$
		HGR	$0.580 \pm .531$	$0.371 \pm .343$	$0.597 \pm .545$
		PER	$\mathbf{0.600 \pm .548}$	$\mathbf{0.406 \pm .374}$	$0.596 \pm .544$
DrawerClose	DDPG	**SABER (Ours)**	$0.280 \pm .251$	$0.324 \pm .082$	$0.277 \pm .100$
		HER	$0.160 \pm .182$	$0.251 \pm .030$	$0.250 \pm .093$
		HGR	$0.080 \pm .091$	$\mathbf{0.349 \pm .184}$	$0.226 \pm .052$
		PER	$\mathbf{0.460 \pm .363}$	$0.308 \pm .049$	$\mathbf{0.340 \pm .063}$
	TD3	**SABER (Ours)**	$0.590 \pm .539$	$0.549 \pm .422$	$0.534 \pm .485$
		HER	$0.350 \pm .480$	$0.329 \pm .348$	$0.284 \pm .365$
		HGR	$0.500 \pm .457$	$\mathbf{0.602 \pm .305}$	$0.460 \pm .381$
		PER	$\mathbf{0.600 \pm .548}$	$0.524 \pm .379$	$\mathbf{0.583 \pm .510}$

Table 1: Primary post-HER comparisons over paired seeds 1–5 (mean $\pm$ sample standard deviation). Final denotes final-checkpoint success; AUC and FQ-AUC denote full-span and final-quarter normalized success AUC. Bold denotes the highest mean within each task–backbone and metric; ties are bolded. All methods include HER, and repeated HER+ prefixes are omitted.

Matched HER comparison. SABER achieves higher mean full-span AUC than matched HER in all seven task–backbone cohorts while preserving replay sampling, HER relabeling, Bellman targets, actor optimization, architecture, and UTD. On FPP, mean AUC increases from 0.539 to 0.609 with DDPG, from 0.372 to 0.442 with TD3, and from 0.471 to 0.583 with SAC. Paired 95% confidence intervals exclude zero for TD3 and SAC but include zero for DDPG. A separately frozen FPP TD3 confirmatory extension to ten paired seeds yields a mean SABER-minus-HER AUC difference of $+0.0467$ (paired-bootstrap 95% CI $[0.0085, 0.0783]$), with SABER higher in 8/10 pairs; seed-level values and the exact sign-flip analysis are reported in the supplementary material. A paired five-seed sensitivity run using an inclusive future-goal convention that also admits $m(s_{t+1})$ yields a mean FPP TD3 gain of $+0.0674$, close to $+0.0708$ under the primary convention (95% CI $[-0.0068, 0.1416]$). This preserves the observed direction and provides evidence that the primary gain is not driven by this indexing choice; full paired results are reported in the supplementary material. The largest full-span gain occurs on DrawerOpen DDPG, where AUC increases from 0.207 to 0.754, final-quarter AUC from 0.393 to 0.987, and final success from 0.390 to 1.000. Mean full-span AUC also increases under both DDPG and TD3 on DrawerClose, and the final-quarter metric retains the same observed SABER-over-HER direction in all seven cohorts. Thus, SABER attained a higher observed mean AUC in all seven five-seed cohorts, while paired interval evidence varied by cohort.

HGR comparison. HGR and SABER address different allocation decisions in HER-based learning: HGR changes which hindsight experiences enter an update, whereas SABER changes the relative critic-fitting contribution of experiences already selected. SABER achieves higher mean full-span AUC on all three FPP backbones and on DrawerOpen DDPG. HGR is slightly higher on DrawerOpen TD3 and higher on both DrawerClose backbones. On DrawerClose, however, SABER has higher observed final-quarter AUC: 0.277 versus 0.226 with DDPG and 0.534 versus 0.460 with TD3. The paired intervals for these late-training differences include zero, so the result indicates a metric-dependent, task-dependent ordering rather than stronger evidence of stability or a universal ranking. In our recorded implementations, HGR has lower throughput than SABER in every evaluated cohort; these hardware-specific measurements are reported separately and are not used as a performance claim.

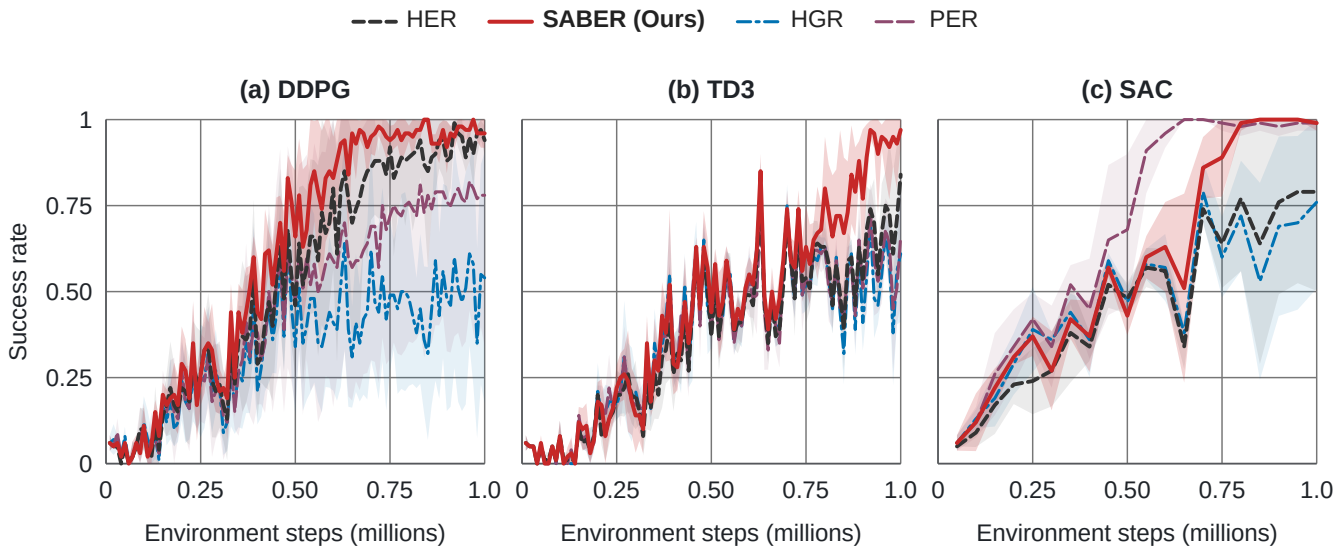


Figure 3: FetchPickAndPlace learning curves for the primary comparisons across DDPG, TD3, and SAC. Lines show mean deterministic evaluation success over paired seeds 1–5, and shaded regions show sample standard deviation. DDPG and TD3 use the exact 10k evaluation grid, while SAC uses the exact shared 50k grid. No smoothing or interpolation is applied; corresponding full-span AUC, final-quarter AUC, and final-success values are reported in Table 1.

PER comparison. Across all seven cohorts, PER produces a mixed ordering. SABER has higher mean full-span AUC on FPP DDPG, FPP TD3, and both DrawerClose backbones, whereas PER is higher on FPP SAC and both DrawerOpen backbones. DrawerClose TD3 also exposes a metric trade-off: SABER attains higher full-span AUC (0.549 versus 0.524), whereas PER has higher final-quarter AUC (0.583 versus 0.534) and marginally higher mean final success (0.600 versus 0.590). PER changes the replay distribution and applies importance correction, so it is a broader replay-selection intervention rather than a matched loss-allocation control. Neither replay prioritization nor critic-loss allocation therefore uniformly dominates across tasks, backbones, and metrics.

4.3 Diagnostic Instrumentation

Diagnostic logging adds no gradient terms and does not alter replay sampling, training objectives, or the parameter-update order. For FPP, we record one full minibatch every 10k environment steps for seeds 1–5 of DDPG, TD3, and SAC, yielding 500 diagnostic batches per backbone and 1,500 in total. DrawerOpen and DrawerClose DDPG use separate ten-seed diagnostic cohorts. Snapshot-level quantities are first averaged within each seed, and seeds are treated as the independent analysis units.

Support activation. We first measure how often SABER's support condition activates during training. The support-positive rate is the fraction of logged samples for which $\psi_i = 1$. Across the five instrumented cohorts, it ranges from 0.197 on FPP SAC to 0.429 on DrawerOpen DDPG. Thus, the indicator consistently identifies a nontrivial subset of each minibatch rather than degenerating to an almost-always or almost-never signal. Complete cohort-level estimates are reported in the supplement.

TD-error enrichment. We next relate support to the pre-update absolute Bellman residual $|\delta_i^{\mathrm{pre}}|$. On FPP, Pearson correlations with the binary indicator are weak (0.116–0.143), whereas support-positive rates in the top TD-error decile are enriched by 1.409×–1.828× over their base rates. DrawerOpen and DrawerClose show support-positive/negative residual ratios of 1.17× and 1.45×, respectively, with the same direction in all ten seeds of each cohort. Support therefore concentrates in high-error regions without reducing to a TD-error ranking.

One-update residual reduction. To test whether support-conditioned weights affect the intended critic update, we hold the Bellman targets fixed and measure $|\delta_i^{\mathrm{pre}}| - |\delta_i^{\mathrm{post}}|$ on the same minibatch after one critic step. Support-positive/negative reduction ratios are 2.39×, 3.30×, and 3.94× for FPP DDPG, TD3, and SAC, and 2.10× and 3.48× for DrawerOpen and DrawerClose. The positive-sample reduction is larger in all 35 backbone–seed runs. These results verify that SABER translates its support signal into the intended redistribution of immediate critic fitting; the independent contribution of exact sample assignment is evaluated in the subsequent ablations.

Phase-conditioned allocation. Finally, we examine where support activates along Drawer task progression using phases conditioned on the relabeled training goal. The completed open and closed phases account for 73.8% and 74.5% of logged samples, respectively, but have the lowest support-

positive rates (0.326 and 0.208). SABER therefore activates more frequently around approach and drawer interaction than in the dominant terminal phase. Full phase definitions, counts, and seed-level results are reported in the supplement.

4.4 Mechanism Ablations

The primary comparisons establish SABER's performance effect but do not identify whether it arises from the action used to query support, the pairing between that action and the sampled context, or the sample-wise placement of high critic weights. We therefore evaluate four DDPG controls on FPP, DrawerOpen, and DrawerClose with paired seeds 1–5. All controls retain Main SABER's task-specific budget, HER and replay procedures, support hyperparameters, critic-only scope, UTD ratio, and evaluation protocol without variant-specific retuning. Table 2 reports paired Main-minus-control AUC differences; positive values favor Main.

Replay-Action Query. Replay-Action Query replaces a_i^π with a_i^r only in the support query, retaining the context, population threshold, weight computation, replay-action regression, and actor objective. Because the sampled transition remains in its local replay pool, its replay action can support itself within the bounded scan. Main is higher on all three tasks, with the interval excluding zero only on DrawerOpen (Table 2), showing that query identity can materially affect learning.

Shuffled-Policy Query. Shuffled-Policy Query applies a fixed-point-free within-minibatch permutation to the current-policy actions, preserving their multiset while breaking sample–query pairing. The shuffled action is used only for the support query, and its weight remains attached to the original replay-action residual. Main is higher on all three tasks, although all intervals include zero. Because Replay-Action and Shuffled-Policy Query can also change the support-positive count, they test the overall effects of query identity and pairing rather than alignment under a matched count.

Random-Support. At each update, Random-Support applies Main SABER's indicator rule to the control run's own minibatch and replay state, obtains a counterfactual mask M with $k = \sum_i M_i$, and assigns exactly k high weights uniformly without replacement across the minibatch. It matches Main's positive count and raw and normalized weight distributions while removing sample-level context eligibility and action-bin membership from the high-weight locations. Main is higher on all three tasks, with the FPP interval excluding zero. Because k still comes from Main's support rule, this is a count-matched control rather than a support-free baseline.

Strict Context-Eligible Count-Matched. This stricter control draws the same k samples without replacement only from the eligible set E satisfying Main SABER's local-context population threshold. It therefore also matches context eligibility, leaving current-policy action-bin membership as the only intended algorithmic difference. Main is higher on all three tasks, but all intervals include zero. The direction is consistent with membership-aligned assignment contributing, but does not independently establish exact support membership as the unique causal mechanism.

Control	FPP	Open	Close
Replay-Action Query	$+0.011$ $[-0.028, 0.059]$	$+0.436$ $[0.121, 0.729]$	$+0.049$ $[-0.006, 0.128]$
Shuffled-Policy Query	$+0.043$ $[-0.002, 0.085]$	$+0.105$ $[-0.080, 0.335]$	$+0.033$ $[-0.063, 0.119]$
Random-Support	$+0.056$ $[0.014, 0.101]$	$+0.047$ $[-0.042, 0.136]$	$+0.048$ $[-0.004, 0.115]$
Strict Count-Matched	$+0.033$ $[-0.006, 0.076]$	$+0.149$ $[-0.042, 0.442]$	$+0.030$ $[-0.033, 0.092]$

Table 2: Paired Main-minus-control normalized success AUC differences (95% paired-bootstrap CIs), using DDPG seeds 1–5. Open and Close denote DrawerOpen and DrawerClose; positive values favor Main SABER.

5 Conclusion and Limitations

We introduced SABER, a critic-only objective that, after replay sampling and HER relabeling, reallocates per-sample Bellman-loss coefficients according to local replay support for the current-policy action. Across all seven primary task–backbone cohorts with five paired seeds, SABER achieved higher mean normalized success AUC than matched HER. HGR and PER remained competitive and were stronger in some cohorts. Together, these results identify post-HER critic-loss allocation as a useful design axis distinct from the matched HER pipeline, rather than a universally superior replacement for prioritization.

Across all diagnostic cohorts, support-positive samples exhibited larger residual reductions after a single critic update, confirming that SABER reallocates critic fitting as intended. Importantly, these performance gains were obtained without changing replay sampling, HER relabeling, Bellman targets, actor objectives, network architectures, or UTD; SABER modifies only the critic-loss coefficients within the existing HER minibatch. SABER therefore demonstrates that a limited change to supervision allocation can materially improve learning without introducing new samples or targets.

Our evidence is limited to three simulated manipulation tasks and five-seed primary cohorts, and matched SAC coverage is unavailable for the Drawer tasks. SABER also relies on task-specific contexts, fixed support thresholds, and implementation-ordered bounded replay retrieval; sensitivity to candidate ordering, including temporally recent retrieval, remains untested. It is not an exploration method that discovers useful actions absent from replay. Current support retrieval incurs additional computation, so equal UTD does not imply equal wall-clock cost. Future work should investigate adaptive support representations, faster retrieval, broader tasks, larger seed cohorts, and resource costs. Within these boundaries, our results show that HER learning can improve without changing the selected or relabeled experience itself.

References

Andrychowicz, M.; Wolski, F.; Ray, A.; Schneider, J.; Fong, R.; Welinder, P.; McGrew, B.; Tobin, J.; Abbeel, P.; and Zaremba, W. 2017. Hindsight Experience Replay. In *Advances in Neural Information Processing Systems*, volume 30, 5048–5058. Curran Associates, Inc.

Chen, X.; Wang, C.; Zhou, Z.; and Ross, K. 2021. Randomized Ensembled Double Q-Learning: Learning Fast Without a Model. In *International Conference on Learning Representations*.

Fang, M.; Zhou, T.; Du, Y.; Han, L.; and Zhang, Z. 2019. Curriculum-Guided Hindsight Experience Replay. In *Advances in Neural Information Processing Systems*, volume 32, 12623–12634. Curran Associates, Inc.

Fujimoto, S.; Meger, D.; and Precup, D. 2019. Off-Policy Deep Reinforcement Learning without Exploration. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, 2052–2062. PMLR.

Fujimoto, S.; Meger, D.; and Precup, D. 2020. An Equivalence between Loss Functions and Non-Uniform Sampling in Experience Replay. In *Advances in Neural Information Processing Systems*, volume 33, 14219–14230. Curran Associates, Inc.

Fujimoto, S.; van Hoof, H.; and Meger, D. 2018. Addressing Function Approximation Error in Actor-Critic Methods. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, 1587–1596. PMLR.

Haarnoja, T.; Zhou, A.; Abbeel, P.; and Levine, S. 2018. Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, 1861–1870. PMLR.

Hiraoka, T. 2023. Efficient Sparse-Reward Goal-Conditioned Reinforcement Learning with a High Replay Ratio and Regularization. arXiv:2312.05787.

Kumar, A.; Fu, J.; Soh, M.; Tucker, G.; and Levine, S. 2019. Stabilizing Off-Policy Q-Learning via Bootstrapping Error Reduction. In *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc.

Kumar, A.; Gupta, A.; and Levine, S. 2020. DisCor: Corrective Feedback in Reinforcement Learning via Distribution Correction. In *Advances in Neural Information Processing Systems*, volume 33, 18560–18572. Curran Associates, Inc.

Kumar, A.; Zhou, A.; Tucker, G.; and Levine, S. 2020. Conservative Q-Learning for Offline Reinforcement Learning. In *Advances in Neural Information Processing Systems*, volume 33, 1179–1191. Curran Associates, Inc.

Lanka, S.; and Wu, T. 2018. ARCHER: Aggressive Rewards to Counter Bias in Hindsight Experience Replay. arXiv:1809.02070.

Levine, A.; and Feizi, S. 2023. Goal-Conditioned Q-learning as Knowledge Distillation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, 8500–8509.

Lillicrap, T. P.; Hunt, J. J.; Pritzel, A.; Heess, N.; Erez, T.; Tassa, Y.; Silver, D.; and Wierstra, D. 2016. Continuous Control with Deep Reinforcement Learning. In *International Conference on Learning Representations*.

Liu, X.-H.; Xue, Z.; Pang, J.; Jiang, S.; Xu, F.; and Yu, Y. 2021. Regret Minimization Experience Replay in Off-Policy Reinforcement Learning. In *Advances in Neural Information Processing Systems*, volume 34, 17604–17615. Curran Associates, Inc.

Luu, T. M.; and Yoo, C. D. 2021. Hindsight Goal Ranking on Replay Buffer for Sparse Reward Environment. *IEEE Access*, 9: 51996–52007.

Novati, G.; and Koumoutsakos, P. 2019. Remember and Forget for Experience Replay. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, 4851–4860. PMLR.

Plappert, M.; Andrychowicz, M.; Ray, A.; McGrew, B.; Baker, B.; Powell, G.; Schneider, J.; Tobin, J.; Chociej, M.; Welinder, P.; Kumar, V.; and Zaremba, W. 2018. Multi-Goal Reinforcement Learning: Challenging Robotics Environments and Request for Research. arXiv:1802.09464.

Ren, Z.; Dong, K.; Zhou, Y.; Liu, Q.; and Peng, J. 2019. Exploration via Hindsight Goal Generation. In *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc.

Schaul, T.; Horgan, D.; Gregor, K.; and Silver, D. 2015. Universal Value Function Approximators. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, 1312–1320. PMLR.

Schaul, T.; Quan, J.; Antonoglou, I.; and Silver, D. 2016. Prioritized Experience Replay. In *International Conference on Learning Representations*.

Schramm, L.; Deng, Y.; Granados, E.; and Boularias, A. 2023. USHER: Unbiased Sampling for Hindsight Experience Replay. In *Proceedings of the 6th Conference on Robot Learning*, volume 205 of *Proceedings of Machine Learning Research*, 2073–2082. PMLR.

Sinha, S.; Song, J.; Garg, A.; and Ermon, S. 2022. Experience Replay with Likelihood-free Importance Weights. In *Proceedings of the 4th Annual Learning for Dynamics and Control Conference*, volume 168 of *Proceedings of Machine Learning Research*, 110–123. PMLR.

Sun, P.; Zhou, W.; and Li, H. 2020. Attentive Experience Replay. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, 5900–5907.

Yang, Y.; Yang, R.; Lyu, J.; Yan, J.; Luo, F.; Luo, D.; Li, X.; and Li, L. 2023. Multi-Step Hindsight Experience Replay with Bias Reduction for Efficient Multi-Goal Reinforcement Learning. In *Proceedings of the 2023 International Conference on Frontiers of Robotics and Software Engineering*, 144–156. IEEE.

Yu, T.; Quillen, D.; He, Z.; Julian, R.; Hausman, K.; Finn, C.; and Levine, S. 2020. Meta-World: A Benchmark and Evaluation for Multi-Task and Meta Reinforcement Learning. In *Proceedings of the Conference on Robot Learning*,

volume 100 of *Proceedings of Machine Learning Research*, 1094–1100. PMLR.

Zhao, R.; and Tresp, V. 2018. Energy-Based Hindsight Experience Prioritization. In *Proceedings of the 2nd Conference on Robot Learning*, volume 87 of *Proceedings of Machine Learning Research*, 113–122. PMLR.

Zheng, S.; Bai, C.; Yang, Z.; and Wang, Z. 2024. How Does Goal Relabeling Improve Sample Efficiency? In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 61246–61266. PMLR.